



Catalyst/Paraview installation

Luís Paulo Santos, Dezembro, 2025

Running *in-situ* visualizations requires the Catalyst Paraview API and library to be installed within your system. Unfortunately, the binary distributions of Paraview usually do not include the Catalyst library.

Please check whether that is the case in your system. Let `PARAVIEW_PATH` hold the path to your Paraview installation. Check whether

`($PARAVIEW_PATH)/lib/catalyst/libcatalyst-paraview.so`

exists. If not, you'll have to follow the steps in sections 1 and 2, below. Otherwise, proceed to section 3.

Section 1 – Build Catalyst

You can find Catalyst 2.0 installation and building files and instructions in

https://catalyst-in-situ.readthedocs.io/en/latest/build_and_install.html

I strongly recommend that you use `ccmake` to visually inspect the building configuration. Below is an image with my settings. Note that I disabled MPI for my laptop installation and I manually set my system architecture to 'arm64'. If you want MPI enabled, make sure you use the same MPI version for Catalyst, Paraview, and, later, for your simulation application.

Then use `ccmake` to configure and generate your build files (using Ninja, make or whatever system you want and CMake supports).

```
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CATALYST_BUILD_REPLAY          ON
CATALYST_BUILD_SHARED_LIBS     ON
CATALYST_BUILD_STUB_IMPLEMENTA ON
CATALYST_BUILD_TESTING         ON
CATALYST_BUILD_TOOLS           ON
CATALYST_USE_MPI                OFF
CATALYST_WRAP_FORTRAN          OFF
CATALYST_WRAP_PYTHON           OFF
CMAKE_BUILD_TYPE                Release
CMAKE_INSTALL_PREFIX            /Users/psantos/MCA/my-catalyst/libcatalyst
CMAKE_OSX_ARCHITECTURES         arm64
CMAKE_OSX_DEPLOYMENT_TARGET
CMAKE_OSX_SYSROOT
DUMPBIN_EXECUTABLE              DUMPBIN_EXECUTABLE-NOTFOUND
GTest_DIR                      GTest_DIR-NOTFOUND
NM_EXECUTABLE                   /usr/bin/nm
```

After building Catalyst your folder `CMAKE_INSTALL_PREFIX` should contain the `bin`, `include` and `lib` folders with the respective files.

In the next section, I will refer to the Catalyst installation directory (currently `CMAKE_INSTALL_PREFIX`) as `CATALYST_DIR`.

Section 2 – Build Paraview

Download the source code of Paraview 6.0 from <https://www.paraview.org/download/>.

Once unzipped you can find the build documentation in

`($PARAVIEW_DOWNLOAD_PATH)/Documentation/dev/build.md`

where `PARAVIEW_DOWNLOAD_PATH` refers to the path where you uncompressed the sources.

To set up the building configuration, use `ccmake`.

In my configuration I was careful to assure that the target architecture is `arm64` (I am using a Mac), the build version is `Release` and to setup `PARAVIEW_USE_MPI` identical to the Catalyst build (i.e., `OFF`).

Ensure that the `PARAVIEW_ENABLE_CATALYST` option is set to `ON`. This is the purpose of this building effort.

Ensure the `catalyst_DIR` option is set to `($CATALYST_DIR)/lib/cmake`, where `ccmake` can find Catalyst's CMake configuration files.

After configuring and generating the build files you can proceed with the build. I used `make`.

ALERT: Building Paraview took several hours on my machine. I suspect, though, that the system was memory trashing.

Check whether you can find the following file:

`($PARAVIEW_PATH)/lib/catalyst/libcatalyst-paraview.so`

Section 3 – Build and execute Catalyst example

Build and run the official Catalyst example, following the instructions available in

https://docs.paraview.org/en/latest/Catalyst/getting_started.html#building-the-example